

Extended Area Blackbody System

SR-800N

COMMUNICATION PROTOCOL DESCRIPTION

©Copyright 2016, CI-Systems

This manual has been published by CI-Systems and supplied on condition that it is used solely for the purpose of supporting the operation, service, and maintenance of CI-Systems equipment. Mutually accepted provisions of the contract with CI-Systems will govern the rights of the customer with respect to this manual. This manual shall not be duplicated, released, disclosed, or used, in whole or in part, for any purpose other than that stated herein, without the express written permission of CI-Systems.

Technical Support

Tel: ++972-4-6448888

Fax: ++972-4-6543570

Address: CI-Systems P.O. Box 147

Industrial Park, Ramat-Gabriel, Migdal-Ha'Emek, Israel 10551

E-mail: cis@ci-systems.com

Web-site: www.ci-systems.com

Preface

Scope of This Manual

This Document provides all the required information for using the communication protocol of the SR-800N Extended Area Blackbody System.

Contents of This Manual

The manual includes the following chapters:

Chapter 1: **Introduction & General information**

Introduction to the SR-800N Extended Area Blackbody and its Communication Protocol.

Chapter 2: **Protocol Description**

Description of the Communication Protocol of the SR-800N, messages structure and codes list

Chapter 3 **Messages Examples**

Examples of messages to the SR-800N, including its response.

Chapter 4 **Index**

List of Updates (Effective Pages)

No.	Date	Description	Page Number
A	October 2016	New Release	All

Table of Contents

1.	INTRODUCTION & GENERAL INFORMATION	1-1
1.1	SR-800N Extended Area Blackbody System Overview	1-1
1.2	SR-800N Communication Protocol Overview	1-2
1.3	SR-800N Communication channels setting	1-3
1.3.1	Serial Communication RS232 setting	1-3
1.3.2	Ethernet	1-3
1.3.3	Bytes order	1-3
1.3.4	Messages Sequence	1-3
2.	PROTOCOL DESCRIPTION	2-4
2.1	Message structure	2-4
2.1.1	General Message structure	2-4
2.1.2	Service Code 'Set Parameters' Message structure	2-5
2.1.3	Service Code 'Get Parameters' Message structure	2-6
2.1.4	Parameters Codes list	2-7
3.	MESSAGES EXAMPLES	3-14
3.1	Set Parameters example (see Section 2.1.2).....	3-14
3.1.1	Blackbody 'Set Operation mode' (see section 2.1.4.2)	3-14
3.1.2	Blackbody 'Set Setpoint (Absolute)' (see section 2.1.4.2).....	3-14
3.2	Get Parameters example (see Section 2.1.2)	3-15
3.2.1	Blackbody 'Get Operation mode' and 'Get current set point' (see section 2.1.4.2).....	3-15
3.2.2	Response to Blackbody 'Get Operation mode' and 'Get current set point' command (see section 2.1.4.2)	3-15
4.	INDEX	4-16

List of Tables

Table 2-1:	General Message structure	2-4
Table 2-2:	Service Code 'Set Parameters' Message structure.....	2-5
Table 2-3:	Service Code 'Get Parameters' Message structure	2-6
Table 2-4:	General Codes	2-7
Table 2-5:	BlackBody Control Codes	2-7
Table 2-6:	BlackBody Configuration Codes	2-8
Table 2-7:	Axes Control Codes	2-9
Table 2-8:	Axes Configuration Codes	2-10
Table 2-9:	'Preset' Control Codes	2-13
Table 2-10:	'I/O' Control Codes.....	2-13

List of Figures

Figure 1-1:	SR-800N Extended Area Blackbodies	1-2
-------------	---	-----

List of Acronyms and Abbreviations

ATE:	Automated Test Equipment
BB:	Blackbody
BIT:	Built In Test
CCW:	Counterclockwise
CW:	Clockwise
EFL:	Effective Focal Length
I/O:	Input Output
IP:	Internet Protocol
IR	Infra-Red
LAN:	Local Array Network
LCD:	Liquid Crystal Display
LSB:	Least Significant Byte
RTD	resistance temperature detectors
VIP:	Version Independent Protocol

Applicable Documents

➤ **CI Systems Documents**

- Doc No. 6057050: SR800N - Installation and operation manual

1. Introduction & General information

This chapter serves as an introduction to the SR-800N Extended Area Blackbody System.

1.1 SR-800N Extended Area Blackbody System Overview

The SR-800N Extended Area Blackbody System consists of a high emissivity and uniformity Blackbody with its controller (the ControlMaster).

The ControlMaster measures the temperature of the Blackbody with high precision to achieve high temperature accuracy and stability.

The Blackbodies are available in various emitter sizes and temperature ranges.

The Blackbodies are equipped with removable temperature sensor (RTD) which enables fast and easy calibration of the Blackbody by replacing the sensor with new calibrated one and updating the calibration parameters, which are stored in the Blackbody head.

The customer may add accessories such as target wheel, Filter wheel etc..

The Blackbody can be operated in absolute mode – according to the temperature of the emitter, or in differential mode – the emitter temperature is relative to temperature of the target in front of it.

SR-800N Extended Area Blackbody System can be operated as a stand-alone system with local operation using the front panel of the ControlMaster, or may be integrated as part of an Electro-optics Test System, with remote operation via the communication ports of the ControlMaster.

Figure 1-1: SR-800N Extended Area Blackbodies



1.2 SR-800N Communication Protocol Overview

The SR-800N Communication Protocol is a VIP (Version-Independent Protocol) protocol which is designed for communication between computers, controllers or other devices.

The advantage of this type of protocol is that it enables those devices to communicate regardless the type of network that connects them.

This document describes the following issues:

- The process which a device uses in order to request an access to another device
- The response of requests from the other devices.
- The structure of a message, with description of its fields.
- List of standard messages, which are used in the SR-800N.
- Example for standard messages.

1.3 SR-800N Communication channels setting

1.3.1 Serial Communication RS232 setting

- Baud rate – 115200 bps
- Start Bits -1
- Stop Bits -1
- No Parity.
- 8 data bits.

1.3.2 Ethernet

- UDP Protocol
- Port 5200
- Maximal message size – 1024 bytes

1.3.3 Bytes order

The VIP protocol uses standard network bytes order - big-endian, LSB first.

1.3.4 Messages Sequence

The Host Computer initiates the communication sequence with the SR-800N.

2. Protocol Description

This chapter describes the structure of the Communication Protocol of the SR-800N.

2.1 Message structure

2.1.1 General Message structure

Table 2-1: General Message structure

[Sync byte][Address ID][Size][Service Code][Data....][Check Sum]

Field Name	Size (Bytes)	Description
Sync Byte	1	Synchronization byte, <u>always 0xAA</u>
Address ID	1	This field is used only when using this protocol with serial communication RS485. (Serial bus with several devices). For all other types of communication (RS232, Ethernet) should be set to <u>1</u> .
Size	2	Message size, not including itself, the Sync byte and address. (i.e only the Data Size + 2 (Service Code and Check Sum fields).
Service Code	1	A number indicating the service request / response / notification is being issued. For example, the code of service 'Send parameters' is 6, 'Read parameters' - 8. The value '0' is invalid.
Data	4-1014	Data block contains chain of parameters. (See sections 2.1.2 and 2.1.3)
Check Sum (CS)	1	The Sum of all the bytes in the message including the CS field must be 0x00. CS byte is equal to 0x00 - sum of all bytes excluding CS byte.

2.1.2 Service Code 'Set Parameters' Message structure

2.1.2.1 Service Code Value

The Service code for 'set Parameters' is: 0x06

2.1.2.2 Message structure

Table 2-2: Service Code 'Set Parameters' Message structure

Field Name	Value	Size (Bytes)	Note
Sync Byte	0xAA	1	
Address ID	0x01	1	
Size		2	Calculated
Service Code	0x06	1	
Data	Parameter1 ID	2	
	Parameter1 Size	2	0-1014
	Parameter1 data	0-1014	
	Parameter2 ID	2	
	Parameter2 Size	2	0-1014
	Parameter2 data	0-1014	
	...		
	ParameterN ID	2	
	ParameterN Size	2	0-1014
	ParameterN data	0-1014	
Check Sum (CS)		1	calculated

Note: setting the size value of a parameter to 0 means that this parameter is unknown in the current version of the system.

2.1.2.3 Response

There is no response for this service request.

2.1.3 Service Code 'Get Parameters' Message structure

2.1.3.1 Service Code Value

The Service code for 'set Parameters' is: 0x08

2.1.3.2 Message structure

Table 2-3: *Service Code 'Get Parameters' Message structure*

Field Name	Value	Size (Bytes)	Note
Sync Byte	0xAA	1	
Address ID	0x01	1	
Size		2	Calculated
Service Code	0x08	1	
Data	Parameter1 ID	2	
	0x0000	2	
	Parameter2 ID	2	
	0x0000	2	
	...		
	ParameterN ID	2	
	0x0000	2	
Check Sum (CS)		1	calculated

2.1.3.3 Response

The response for this service request is a 'Set Parameter service' (see [Section 2.1.2](#)) from the SR-800N.

2.1.4 Parameters Codes list

The following table describes the available parameters codes.

2.1.4.1 General Codes

Table 2-4: General Codes

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Hardware Type	0x0064	100	x		R	4	Integer	0x210
Serial Number	0x0065	101	x		R	20	Char	
Software Version	0x0066	102	x		R	20	Char	
Control Master Model	0x177A	6010	x		R	20	Char	

2.1.4.2 BlackBody Control Codes

Table 2-5: BlackBody Control Codes

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Temperature Is Stable	0x07D5	2005	x		R	1	Byte	0 - not stable 1- stable
Current temperature	0x07D7	2007	x		R	4	Float	Absolute or Differential
Emitter Temperature	0x07D8	2008	x		R	4	Float	
Target Temperature	0x07D9	2009	x		R	4	Float	
Differential Temperature	0x07DA	2010	x		R	4	Float	
Operation Mode	0x07F0	2032	x	x	RW	1	Byte	1- Absolute 2- Differential
Set Point (Absolute)	0x07F1	2033	x	x	RW	4	Float	
Set Point (Differential)	0x07F2	2034	x	x	RW	4	Float	
Current Set Point	0x07F3	2035	x		R	4	Float	Absolute or Differential

2.1.4.3 BlackBody Configuration Codes

Table 2-6: BlackBody Configuration Codes

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
BlackBody S/N	0x07EF	2031	x		R	20	Char	
Radiometric Offset Value	0x07F5	2037	x	x	RW	4	Float	
Radiometric Offset Enabled	0x07F6	2038	x	x	RW	1	Boolean	True – enabled False - disabled
BlackBody Model	0x07F8	2040	x		R	20	Char	
Operation Mode Options	0x0803	2051	x		R	1	Byte	0- Abs. & Diff 1- Abs. only 2- Diff. Only
BlackBody Emitter Sensor S/N	0x0808	2056	x		R	10	Char	
BlackBody Target Sensor S/N	0x0809	2057	x		R	10	Char	
Absolute Set Point Minimum	0x0834	2100	x		R	4	Float	
Absolute Set Point Maximum	0x0835	2101	x		R	4	Float	
Differential Set Point Minimum	0x0836	2102	x		R	4	Float	
Differential Set Point Maximum	0x0837	2103	x		R	4	Float	

2.1.4.4 Axes Control Codes

Table 2-7: Axes Control Codes

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Axis 1 Status	0x1F41	8001	x		R	1	Byte	0 - stable 1 - moving
Axis 1 Current Position Index	0x1F43	8003	x		R	2	Word	zero based
Axis 1 Home command	0x1F6A	8042		x	W	1	Byte	Data > 0
Axis 1 Position Index Command	0x1F70	8048		x	W	2	Word	zero based
Axis 1 Homming Command	0x1F71	8049		x	W	1	Byte	Data > 0
Axis 2 Status	0x2329	9001	x		R	1	Byte	0 - stable 1 - moving
Axis 2 Current Position Index	0x232B	9003	x		R	2	Word	zero based
Axis 2 Home command	0x2352	9042		x	W	1	Byte	Data > 0
Axis 2 Position Index Command	0x2358	9048		x	W	2	Word	zero based
Axis 2 Homming Command	0x2359	9049		x	W	1	Byte	Data > 0
Axis 3 Status	0x2711	10001	x		R	1	Byte	0 - stable 1 - moving
Axis 3 Current Position Index	0x2713	10003	x		R	2	Word	zero based
Axis 3 Home command	0x273A	10042		x	W	1	Byte	Data > 0
Axis 3 Position Index Command	0x2740	10048		x	W	2	Word	zero based
Axis 3 Homming Command	0x2741	10049		x	W	1	Byte	Data > 0
Axis 4 Status	0x2AF9	11001	x		R	1	Byte	0 - stable 1 - moving
Axis 4 Current Position Index	0x2AFB	11003	x		R	2	Word	zero based
Axis 4 Home command	0x2B22	11042		x	W	1	Byte	Data > 0
Axis 4 Position Index Command	0x2B28	11048		x	W	2	Word	zero based
Axis 4 Homming Command	0x2B29	11049		x	W	1	Byte	Data > 0

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Axis 1 ENC Status	0x1DB1	7601	x		R	1	Byte	0 - stable 1 - moving
Axis 1 ENC Current Index Position	0x1DB3	7603	x		R	2	Word	zero based
Axis 1 ENC Home command	0x1DDA	7642		x	W	1	Byte	Data > 0
Axis 1 ENC Position Index command	0x1DE0	7648		x	W	2	Word	Data > 0
Axis 1 ENC Encoder Position	0x1DF9	7673	x		R	4	Float	
Axis 2 ENC Status	0x1E15	7701	x		R	1	Byte	0 - stable 1 - moving
Axis 2 ENC Current Index Position	0x1E13	7703	x		R	2	Word	zero based
Axis 2 ENC Home command	0x1E3E	7742		x	W	1	Byte	Data > 0
Axis 2 ENC Position Index command	0x1E44	7748		x	W	2	Word	Data > 0
Axis 2 ENC Encoder Position	0x1E5D	7773	x		R	4	Float	

2.1.4.5 Axes Configuration Codes

Table 2-8: Axes Configuration Codes

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Effective Focal Length (EFL)	0x1B60	7008	x	x	RW	4	float	Im mm
Axis 1 Name	0x1F4B	8011	x		R	20	Char	
Axis 1 Installed	0x1F4F	8015	x		R	1	Boolean	True – Installed
Axis 1 Position Indexes Number	0x1F5D	8029	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data
Axis 1 Position Index N Size	0x20AB + 0x000A * N	8363 + 10 * N	x		R	4	float	Size in mm
Axis 1 Position Index N Name	0x20AF + 0x000A * N	8367 + 10 * N	x		R	20	Char	

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Axis 2 Name	0x2333	9011	x		R	20	Char	
Axis 2 Installed	0x2337	9015	x		R	1	Boolean	True – Installed
Axis 2 Position Indexes Number	0x2345	9029	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data
Axis 2 Position Index N Size	0x2493 + 0x000A* N	9363+ 10 * N	x		R	4	float	Size in mm
Axis 2 Position Index N Name	0x2497 + 0x000A* N	9367+ 10 * N	x		R	20	Char	
Axis 3 Name	0x271B	10011	x		R	20	Char	
Axis 3 Installed	0x271F	10015	x		R	1	Boolean	True – Installed
Axis 3 Position Indexes Number	0x272D	10029	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data
Axis 3 Position Index N Size	0x287B + 0x000A* N	10363+ 10 * N	x		R	4	float	Size in mm
Axis 3 Position Index N Name	0x287F + 0x000A* N	10367+ 10 * N	x		R	20	Char	
Axis 4 Name	0x271B	10011	x		R	20	Char	
Axis 4 Installed	0x271F	11015	x		R	1	Boolean	True – Installed
Axis 4 Position Indexes Number	0x272D	11029	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data
Axis 4 Position Index N Size	0x287B + 0x000A* N	11363+ 10 * N	x		R	4	float	Size in mm
Axis 4 Position Index N Name	0x287F + 0x000A* N	11367+ 10 * N	x		R	20	Char	
Axis 1 ENC Name	0x1DBB	7611	x		R	20	Char	
Axis 1 ENC Installed	0x1DBF	7615	x		R	1	Boolean	True – Installed
Axis 1 ENC Position Indexes Number	0x1DCD	7629	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Axis 1 ENC Position Index N Size	0x3433 + 0x000A* N	13363+ 10 * N	x		R	4	float	Size in mm
Axis 1 ENC Position Index N Name	0x3437 + 0x000A* N	13367+ 10 * N	x		R	20	Char	
Axis 2 ENC Name	0x1E1F	7711	x		R	20	Char	
Axis 2 ENC Installed	0x1E23	7715	x		R	1	Boolean	True – Installed
Axis 2 ENC Position Indexes Number	0x1E31	7729	x		R	4	Integer	LSW - Number of Position Indexes MSW – internal data
Axis 2 ENC Position Index N Size	0x381B + 0x000A* N	14363+ 10 * N	x		R	4	float	Size in mm
Axis 2 ENC Position Index N Name	0x381F + 0x000A* N	14367+ 10 * N	x		R	20	Char	

Notes:

(*) The Position Index '**N**' is a zero based number, limited with the number of available Position Indexes for the specified Axis.

(**) The Name of the Position Index is generated automatically from its Type, accompanied with its frequency or Angular size, which is derived from the values of the EFL and its size:

$$\text{Frequency [cy/mrad]} = \text{EFL}/(2000*\text{Size})$$

$$\text{Angular Size [mrad]} = (1000*\text{Size})/\text{EFL}$$

If EFL = 0, the Name contains the Type and the Size in [mm].

If the Size=0, the name contains only the Type.

2.1.4.6 'Preset' Control Codes**Table 2-9: 'Preset' Control Codes**

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
Activate a 'Preset'	0x1772	6002		x	W	4	Integer	The Data contains the number of Preset that will be activate, zero based.

2.1.4.7 'I/O' Control Codes**Table 2-10: 'I/O' Control Codes**

Parameter	Code (hex/Dec)		Get	Set	Access	Data Size	Data Type	Note
I/O type TTL Status	0x1BBD	7101	x	x	RW	1	Byte	Bit 0 – Input 0 Bit 1 – Input 1 Bit 4 – Output 0 Bit 5 – Output 1
I/O type Opto Couplers Status	0x1BBE	7102	x	x	RW	1	Byte	Bit 0 – Input 0 Bit 1 – Input 1 Bit 4 – Output 0 Bit 5 – Output 1
I/O type Open Collector Status	0x1BBF	7103	x	x	RW	1	Byte	Bit 0 – Output 0 Bit 1 – Output 1
I/O type Opto Couplers Status	0x1BC0	7104	x	x	RW	1	Byte	Bit 0 – Output 0 Bit 1 – Output 1

3. Messages Examples

3.1 Set Parameters example (see Section 2.1.2)

3.1.1 Blackbody 'Set Operation mode' (see section 2.1.4.2)

Byte	Value (hex)	Description	Note
0	0xAA	Sync Byte	
1	0x01	Address ID	
2	0x00	Size: 7 bytes	
3	0x07		
4	0x06	Service code 'Send parameter'	
5	0x07	Parameter 1 ID: 0x07F0	
6	0xF0	Blackbody Operation mode	
7	0x00	Parameter 1 Size : 1 byte	
8	0x01		
9	0x01	Parameter 1 Data: 1 (ABS)	
10	0x4F	Check Sum	

3.1.2 Blackbody 'Set Setpoint (Absolute)' (see section 2.1.4.2)

Byte	Value (hex)	Description	Note
0	0xAA	Sync Byte	
1	0x01	Address ID	
2	0x00	Size: 10 bytes	
3	0x0A		
4	0x06	Service code 'Send parameter'	
5	0x07	Parameter 1 ID: 0x07F0	
6	0xF1	Blackbody Setpoint (absolute)	
7	0x00	Parameter 1 Size : 4 byte	
8	0x04		
9	0x42	Parameter 1 Data: 0x42C80000 (floating point, big endian, acc. IEEE 754) = 100.0	
10	0xC8		
11	0x00		
12	0x00		
13	0x3F	Check Sum	

3.2 Get Parameters example (see Section 2.1.2)

3.2.1 Blackbody 'Get Operation mode' and 'Get current set point' (see section 2.1.4.2)

Byte	Value (hex)	Description	Note
0	0xAA	Sync Byte	
1	0x01	Address ID	
2	0x00	Size: 10 bytes	
3	0x0A		
4	0x08	Service code 'Get parameter'	
5	0x07	Parameter 1 ID: 0x07F0	
6	0xF0	Blackbody Operation mode	
7	0x00	Parameter 1 Size : 0 byte (no data)	
8	0x00		
9	0x07	Parameter 2 ID: 0x07D7	
10	0xF0	Blackbody Current temperature	
11	0x00	Parameter 2 Size : 0 byte (no data)	
12	0x00		
13	0x4F	Check Sum	

3.2.2 Response to Blackbody 'Get Operation mode' and 'Get current set point' command (see section 2.1.4.2)

Byte	Value (hex)	Description	Note
0	0xAA	Sync Byte	
1	0x01	Address ID	
2	0x00	Size: 15 bytes	
3	0x0F		
4	0x06	Service code 'send parameter'	
5	0x07	Parameter 1 ID: 0x07F0	
6	0xF0	Blackbody Operation mode	
7	0x00	Parameter 1 Size : 1 byte	
8	0x01		
9	0x01	Parameter 1 data : 1 (ABS)	
10	0x07	Parameter 2 ID: 0x07D7	
11	0xF0	Blackbody Current temperature	
12	0x00	Parameter 2 Size : 4 byte	
13	0x01		
14	0x42	Parameter 2 Data: 0x42C80000 (floating point, big endian, acc. IEEE 754) = 100.0	
15	0xC8		
16	0x00		
17	0x00		
14	0x5B	Check Sum	

4. Index

This chapter provides, in alphabetical order, a list of keywords to be found throughout this manual.

A

Abbreviations · vii

Applicable Documents · vii

C

Codes list

 Axes Configuration · 2-10

 Axes Control · 2-9

 BlackBody Configuration · 2-8

 BlackBody Control · 2-7

 General · 2-7

 'I/O' Control · 2-13

 'Preset' Control · 2-13

Copyright · ii

E

Effective Pages · iv

F

Features · 1-1

G

General Information · 1-1

 Overview · 1-1

I

Introduction · 1-1

L

List of Figures · vi

List of Tables · vi

M

Message structure

 Codes list · 2-7

 General Message · 2-4

 Service Code 'Get Parameters' · 2-6

 Service Code 'Set Parameters' · 2-5

O

Overview · 1-1

P

Preface · iii

Protocol Description · 2-4

 Message structure · 2-4

T

Table of Contents · v

Technical Support · ii